Program : **Diploma in Textile Technology**	
Course Code : **4067**	Course Title: **Yarn manufacture II Lab**
Semester : **4**	Credits: **1.5**
Course Category: **Program core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To impart practical knowledge of the material passage in the speed frame, ring spinning and doubling machines,
- To equip the knowledge of gearing diagram, production, draft and twist calculation

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I	1
Knowledge of spinning preparatory machine		Yarn manufacture I	3
Practical aspects of spinning preparatory machine		Yarn manufacture I Practical	3

Course Outcomes:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the material passage & gearing diagram of speed frame and carry out relevant calculations.	13	Applying
CO2	Demonstrate the material passage & gearing diagram of Ring frame and carry out relevant calculations.	13	Applying
CO3	Demonstrate the material passage of Rotor spinning machine and calculate the production and efficiency.	6	Applying
CO4	Demonstrate the material passage of Ring doubler, TFO machine and calculate the production and efficiency.	10	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Demonstrate the material passage & gearing diagram of speed frame and carry out relevant calculations.**		
M1.01	Study of the material passage of Simplex machine and identify the parts.	3	Understanding
M1.02	Draw the gearing diagram of Speed frame machine and identify change points	3	Applying
M1.03	Calculate draft constant, twist constant and change wheel required for different hank with the help of gearing diagram.	4	Applying
M1.04	Study of drafting roller setting and process parameters for a given roving hank	3	Understanding
CO2	**Demonstrate the material passage & gearing diagram of Ring frame and carry out relevant calculations.**		
M2.01	Study of the material passage of Ring frame machine and identify the parts.	3	Understanding
M2.02	Draw the gearing diagram of Ring frame machine and identify change points	3	Understanding
M2.03	Calculate draft constant, twist constant, change wheel required for different hank with the help of gearing diagram	4	Applying
M2.04	Study of drafting roller setting and process parameters for a particular count, Setting spindle tape drive for s and z twist	3	Understanding
	Lab Exam- I	1	

CO3	**Demonstrate the material passage of Rotor spinning machine and calculate the production and efficiency.**		
M3.01	Study of the material passage of Rotor spinning machine and identify the parts.	3	Understanding
M3.02	Find out production and efficiency of Rotor spinning machine	3	Applying
CO4	**Demonstrate the material passage of Ring doubler, TFO machine and calculate the production and efficiency.**		
M4.01	Study of the material passage of Ring doubler machine and identify the parts.	3	Understanding
M4.02	Study of the material passage of TFO machine and identify the parts.	3	Understanding
M4.03	Estimate the resultant count, production and efficiency of Ring doubler and TFO.	4	Applying
	Lab Exam - II	1	

The students should keep separate record for Yarn manufacture and submit certified & bonafide record at the time of practical examination

Text / Reference:

T/R	Book Title/Author
T1	Lab manual
R2	R. Gokarneshan, Mechanics and calculations of Textile Machinery
R3	W.S. Taggart , Cotton spinning calculation
R4	Heinz Ernst, The Rieter Manual of Spinning, Vol.5 – Rotor spinning, Rieter Machine Works Limited, Switzerland,2014.
R5	T.K.Pattabhiram- Essential calculations of Practical Cotton spinning

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in
2	www.reiter.com
3	www.lmwtmd.com